The unique bimodal restricted fractional-partition sequence

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Task 1. Full solution

For an integer $j > 2$, let $f_j(k)$ count subsets of $\{1, 3, \dots, 2j-1\}$ whose sum is kj, where $1 \le k \le j-1$. Thus $f_j = f_{O_j}$ in Hoelscher–Palsson [1], and their polynomial is $r_j(x) = \sum_{k=1}^{j-1} f_j(k)x^k$. Conjecture 1.0.4 asks whether its coefficient sequence is always unimodal or bimodal. Here unimodality permits a plateau at the maximum, as does the source's discussion of $j = 10$ immediately before the conjecture. Two separated peaks, with a smaller entry between them, constitute bimodality.

Theorem 1. *For every integer $j > 2$ other than 6, the sequence $(f_j(1), \dots, f_j(j-1))$ is unimodal. For $j = 6$ it is $(1, 3, 2, 3, 1)$ and is bimodal. Except at $j = 6, 10$, the sequence increases strictly to its center and decreases strictly thereafter, allowing two equal central entries when j is odd. At $j = 10$ its three central entries are all 20.*

The known theorem used. Write

$$A_j(q) = \prod_{i=1}^{j}(1 + q^{2i-1}) = \sum_{n=0}^{j^2} a_j(n)q^n.$$

Pak–Panova [2, Theorem 5.2] prove that for $j \ge 27$ the sequence $(a_j(26), \dots, a_j(j^2 - 26))$ is symmetric and strictly unimodal, with the usual equal central pair when its length is even. This result supplies the infinite range of our proof. We claim a corollary for the sampled sequence and an exact finite classification, not a new proof of their theorem.

Proof for $j \ge 27$. Choosing a term from each factor of A_j gives $f_j(k) = a_j(kj)$. Complementation of subsets gives

$$a_j(n) = a_j(j^2 - n), \qquad f_j(k) = f_j(j - k). \tag{1}$$

All sampled indices satisfy $26 \le j \le kj \le j^2 - j \le j^2 - 26$. If $1 \le k < \lfloor j/2 \rfloor$, then $kj < (k+1)j \le j^2/2$, so Pak–Panova's theorem gives $f_j(k) < f_j(k+1)$. When j is odd, the two central sampled entries are equal by (1). The remaining inequalities follow by reflection. This proves the assertion in the infinite range. □

Exact certificate for $3 \le j \le 26$. Set $a_0(0) = 1$, set $a_0(n) = 0$ for $n \neq 0$, and interpret out-of-range coefficients as zero. Multiplication by the last factor gives

$$a_j(n) = a_{j-1}(n) + a_{j-1}(n - (2j-1)). \tag{2}$$

For $j \ge 4$ put $d_j = \min_{1 \le k < \lfloor j/2 \rfloor}(a_j((k+1)j) - a_j(kj))$. The finite values are:

j	d_j	j	d_j	j	d_j
3	—	11	6	19	31
4	1	12	8	20	39
5	1	13	9	21	44
6	−1	14	13	22	55
7	2	15	14	23	63
8	1	16	18	24	76
9	3	17	21	25	86
10	0	18	28	26	105

For completeness, the entire computation of this table is the following integer-only implementation of (2). Descending indices ensure that each update uses the previous polynomial's coefficients.

```
a = [1]
for j in range(1, 27):
    w = 2*j - 1
    a += [0]*w
    for n in range(len(a)-1, w-1, -1):
        a[n] += a[n-w]
    if j >= 3:
        half = [a[k*j] for k in range(1, j//2+1)]
        d = min((v-u for u,v in zip(half, half[1:])),
                default=None)
        print(j, d)
```

The certificate file `verification-results.json` also records every sampled coefficient and every first-half difference. The separate verifier checks every coefficient of all 24 polynomials against direct enumeration of the subsets of two halves and convolution of their sum histograms; it uses no floating-point arithmetic.

Completion of the proof. Each positive d_j, together with (1), gives precisely the claimed strict inequalities. The three other rows are, in full,

3	$(1, 1)$
6	$(1, 3, 2, 3, 1)$
10	$(2, 7, 12, 20, 20, 20, 12, 7, 2)$.

They have the asserted shapes. These cases and the preceding infinite argument cover every integer $j > 2$, proving Theorem 1 and Conjecture 1.0.4. □

Scope and attribution. The primary preprint and its 2020 journal final, and Hoelscher's 2021 continuation [3], retain this conjecture. The latter writes the same sampled sequence as Conjecture 1.5. The essential infinite-range input predates both conjectures: Pak–Panova's 2014 Theorem 5.2, building on Almkvist's work. The present deduction consists of restricting that theorem to multiples of j and completing all smaller cases. The source authors already exhibited the $j = 6$ row; its uniqueness is established here. Bounded source and prior-claim checks are recorded separately. The proof and certificate were reviewed by the authoring agent; they have not received independent peer review or proof-assistant certification.

Task 2. A direct injection

Construct an explicit injection from subsets with sum kj into subsets with sum $(k+1)j$ for $k < \lfloor j/2 \rfloor$, outside the two exceptional denominators. Can its missing images explain the positive differences without using the analytic input behind Pak–Panova's theorem?

Task 3. Log-concavity of the sampled rows

Determine for which j the inequalities $f_j(k)^2 \geq f_j(k-1)f_j(k+1)$ hold at every interior index. If they hold for all sufficiently large j, seek an explicit threshold and an exact classification of the smaller cases.

Task 4. Other arithmetic progressions

Replace the odd numerators by $a, a+b, \ldots, a+(j-1)b$ and sample the subset-sum polynomial at multiples of j. Determine which fixed positive integers a, b give eventual unimodality of the resulting nonzero coefficient sequence, and which retain multiple separated peaks.

Task 5. Quantitative gaps and central asymptotics

Obtain explicit estimates for $f_j(k+1) - f_j(k)$ before the center, including the smallest positive gap. Seek uniform local asymptotics near $k = j/2$ and compare their error terms with the exact strict inequalities proved here.

References

[1] Z. Hoelscher and E. Palsson, *Counting restricted partitions of integers into fractions: symmetry and modes of the generating function and a connection to $\omega(t)$*, PUMP J. Undergrad. Res. **3** (2020), 277–307. doi:10.46787/pump.v3i0.2428; arXiv:2011.14502v1.

[2] I. Pak and G. Panova, *Unimodality via Kronecker products*, J. Algebraic Combin. **40** (2014), 1103–1120. doi:10.1007/s10801-014-0520-y; arXiv:1304.5044v2, Theorem 5.2, pp. 7–8 of the author manuscript.

[3] Z. Hoelscher, *Semicomplete arithmetic sequences, division of hypercubes, and the Pell constant*, PUMP J. Undergrad. Res. **4** (2021), 108–116. doi:10.46787/pump.v4i0.2524; arXiv:2102.07083v1.